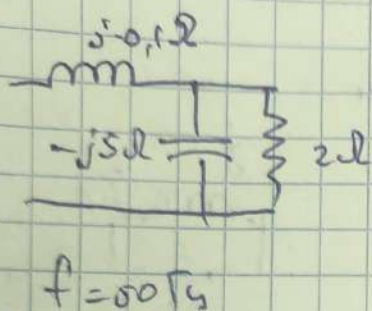


(N1)



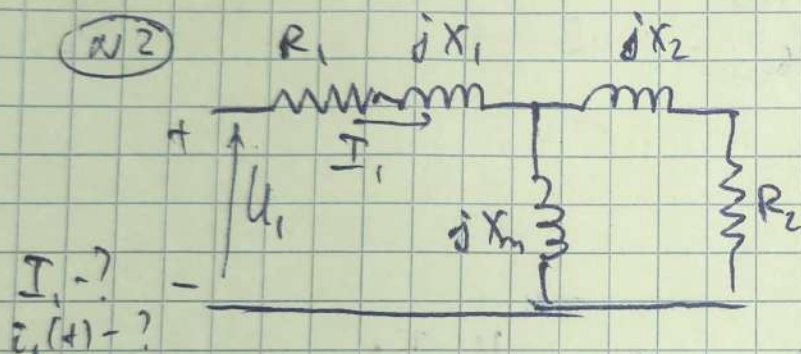
$$Z_2 = \frac{2\Omega \cdot -j50\Omega}{2\Omega - j50\Omega} = \frac{210j}{2 - 5j} =$$

$$= \frac{10j(2 + 5j)}{(2 - 5j)(2 + 5j)} = \frac{50j - 20}{29} \approx$$

$$\approx 1,724 - 0,69j$$

$$Z_2 = Z_1 + 0,1j \Rightarrow Z_{050} = 1,724 - 0,59j$$

(N2)



$U_{1, \text{RMS}} = 220 \text{ V}, f = 50 \text{ Hz}$
 $R_1 = 0,3 \Omega$
 $R_2 = 7,0 \Omega$
 $jX_1 = j0,5 \Omega$
 $jX_2 = j0,2 \Omega$
 $jX_m = j15 \Omega$

правая ветвь: $Z_2 = R_2 + jX_2 = 7 + j0,2$

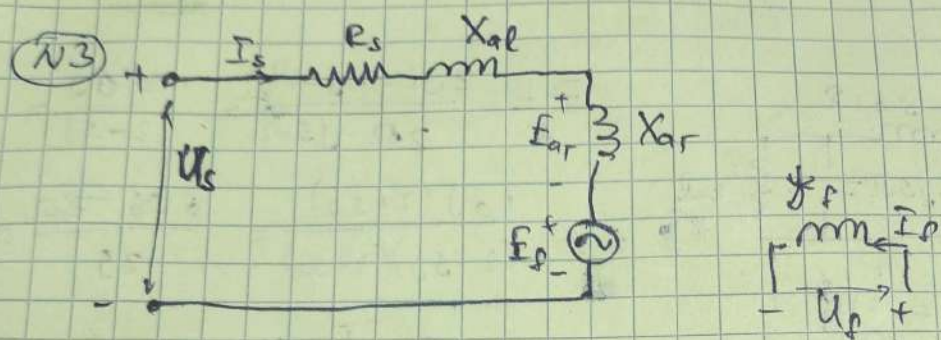
весь правый контур $Z_{2m} = \frac{Z_m Z_2}{Z_m + Z_2} = \frac{j15(7 + j0,2)}{j15 + 7 + j0,2} =$
 $\approx 5,624 + j2,787$

$$Z_{050} = R_1 + jX_1 + Z_{2m} \approx 5,924 + j3,287$$

$$I_1 = \frac{U_1}{Z_{050}} = \frac{220}{5,924 + j3,287} \text{ A} = 28,39 - j15,76 \text{ A}$$

т.е. $|I_1| = 32,47 \quad \varphi_I = 29,0^\circ$

мгновенный ток $i_1(t) = \sqrt{2} \cdot 32,47 \cdot \cos(2\pi \cdot 50t - 29^\circ)$
 $\approx 45,92 \cos(314t - 29^\circ) \text{ A}$



$$p=2$$

$$\omega = 3000 \text{ rpm}$$

$$P_{out} = 10 \text{ kW}$$

$$L_s = 30 \text{ mH}$$

$$R_s = 0,1 \text{ }\Omega$$

$$U_{rms} = 220 \text{ V}$$

$$U_{E_f} = 150 \text{ V}$$

$$f = \frac{p \cdot n}{60} = 50 \text{ Hz}$$

$$X_s = \omega L_s = 2\pi f L_s =$$

$$U_s = E_f + (R_s + jX_s)I_s \Rightarrow I_s = \frac{U_s - E_f}{R_s + jX_s}$$

$$P_{out} = 3 \operatorname{Re}(E_f I_s^*) = 3 \frac{U E_f X_s \sin \delta + U E_f R_s \cos \delta - E_f^2 R_s}{R_s^2 + X_s^2}$$

t.h. $U_s = 220^\circ$ и $\angle = 0$, $E_f = 150^\circ$ и $\angle = \delta$ - угол нагрузки

$$P_{out} = 10 \text{ kW} \rightarrow \frac{\delta_1 = 73^\circ}{\text{без потерь}} \quad \delta_2 = 105,8^\circ$$

$$I_s = \frac{220 - 150 \angle -73^\circ}{0,1 + j9,425} = 15,42 - j18,52 \text{ A}$$

$$|I_s| = 24,1 \text{ A} \quad \varphi_I = -50,2^\circ$$